ModelIt!

Student Activity Pack

How Self-Driving Cars See the Road

Modeling the Sensor Fusion and Decision Systems of Autonomous Vehicles

Grade 10L3 | NGSS HS-PS4-2, HS-PS4-5 | Systems and System Models

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 How do you think a self-driving car knows what is around it without a human watching the road?

2 What sensors do you think a self-driving car uses and what does each one detect?

3 Draw what you think a self-driving car 'sees' compared to what a human driver sees.

4 Why do you think self-driving cars sometimes have accidents even though computers are faster than humans?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

LIDAR Pulse Frequency | Detection Range | Signal Processing Speed | Obstacle Classification | Weather Interference | Decision Latency | Sensor Fusion Accuracy | Route Optimization | Safety Margin

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Clear Highway

Minimal weather interference, highway driving

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Heavy Rain Urban

Heavy rain, complex urban intersection

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Edge Case

Unusual obstacle not in training data

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

V2X Communication — Vehicle-to-Everything communication that allows the car to receive data from other vehicles, traffic signals, and road infrastructure — providing information about hazards beyond sensor range

Map Precision — The accuracy of the pre-built high-definition map the vehicle uses for localization — centimeter-precise maps provide reliable reference but must be continuously updated

Cybersecurity Resilience — The system's ability to resist hacking, sensor spoofing, and data manipulation — a critical safety concern when lives depend on the accuracy of sensor data

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How does a car with no human driver detect, identify, and react to dangers faster than any human — and why does it still sometimes fail? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate why sensor fusion is more reliable than any single sensor type?**

2. **What does the model reveal about the trade-off between Safety Margin and driving efficiency?**

3. **Why are edge cases the fundamental challenge for autonomous vehicles, and can they ever be fully solved?**

4. **How would adding V2X communication change the system's performance in scenarios where sensors are degraded?**

5. **What ethical questions arise from the Safety Margin trade-off — who decides how much risk is acceptable?**

8 STEM Challenge: Design a Self-Driving Sensor Suite for Extreme Weather

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YOUR ENGINEERING MISSION

Design a sensor fusion architecture that maintains safe autonomous driving in severe weather conditions where individual sensor types degrade by 40-70%.

THE SCENARIO

An autonomous vehicle company wants to expand operations to cities with heavy rain, snow, and fog. Current sensor suites lose 40-70% of capability in these conditions. Your team must design a sensor fusion architecture that maintains sufficient Detection Range, Obstacle Classification accuracy, and Safety Margin to operate safely in the worst weather conditions — or determine the threshold at which the car must hand control back to a human driver.

Engineering Design Thinking

1. WHAT SENSOR TYPES AND QUANTITIES DOES YOUR SUITE INCLUDE AND WHERE ARE THEY MOUNTED?
2. HOW DOES YOUR FUSION ALGORITHM ADJUST SENSOR WEIGHTING WHEN SPECIFIC SENSORS ARE DEGRADED?
3. WHAT IS THE MINIMUM DETECTION RANGE AND OBSTACLE CLASSIFICATION ACCURACY YOUR SYSTEM REQUIRES FOR SAFE OPERATION?
4. WHAT IS YOUR HANDOFF PROTOCOL WHEN WEATHER DEGRADES BEYOND SAFE AUTONOMOUS LIMITS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Autonomous Vehicle Engineers design the sensor, perception, and decision systems that enable self-driving cars, trucks, and delivery robots. They work at Waymo, Cruise, Tesla, Aurora, and Zoox, earning $130,000-$300,000/year. This field combines electrical engineering, computer science, and robotics into one of the most cutting-edge career paths in technology.